WASP-135b

R-0199

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-135_250823` · created 2026-07-16T22:14:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460910.776 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.051
Transit depth (Rp/Rs)^2	3.07 +/- 1.78 [%]
Orbital Inclination (inc)	86.98 +/- 2.94
Airmass coefficient 1 (a1)	1.05 +/- 0.78
Airmass coefficient 2 (a2)	-0.0 +/- 0.34
Scatter in the residuals of the lightcurve fit is	76.99 %
Best Comparison Star	#1 - [493, 141]
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.0882 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.051 vs 0.139 (deviation 0.71σ)
Duration fit vs published	0.0882 d vs 0.0687 d (deviation 2.14σ)
Mid-time O–C	-3.8 min
Depth SNR	3.4
Residual scatter	76.99 %

Light curve

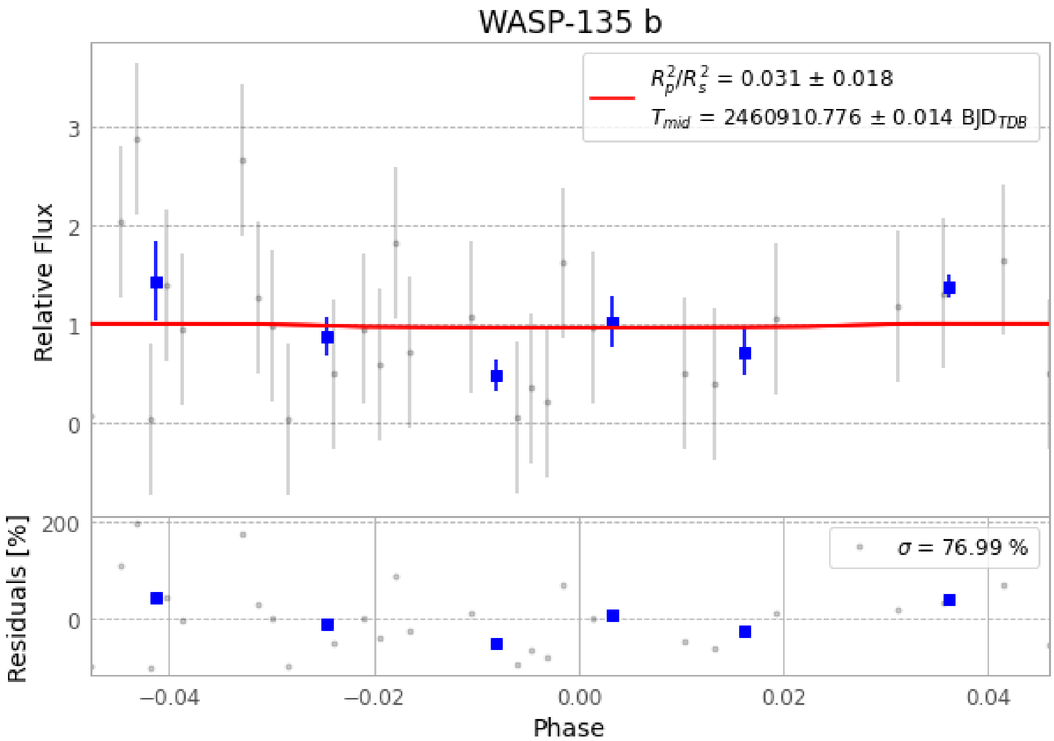

Figure 1 — FinalLightCurve_WASP-135 b_2025-08-22.png (SHA-256 d32308fc912faa21…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 248], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3890c3991ae39b17…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e`

Data provenance

63 FITS frames (41 MB), `WASP-135250823045718.FITS` → `WASP-135250823080615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250823.log (8a21d853382a…), FinalLightCurve_WASP-135 b_2025-08-22.png (d32308fc912f…), FinalLightCurve_WASP-135 b_2025-08-22.csv (c6ea29378a79…)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 22:14Z.